20.4: The practical applications of DNA-based biotechnology affect our lives in many ways

In the last section, we'll survey the practical applications of DNA-based **biotechnology**, the manipulation of organisms or their components to make useful products. Today, major applications of DNA technology and genetic engineering include medicine, forensic evidence and genetic profiles, environmental cleanup, and agriculture.

Medical Applications

One important use of DNA technology is the identification of human genes whose mutation plays a role in genetic diseases. These discoveries may lead to ways of diagnosing, treating, and even preventing such conditions. DNA technology has also identified genes that play a role in a number of "nongenetic" diseases, from arthritis to AIDS, by influencing susceptibility to these diseases. Furthermore, a wide variety of diseases involve changes in gene expression within the affected cells and often within the patient's immune system. By using RNA-seq and DNA microarray assays (see Figures 20.12 and 20.13) or other techniques to compare gene expression in healthy and diseased tissues, researchers are finding genes that are turned on or off in particular diseases. These genes and their products are potential targets for prevention or therapy.

Diagnosis and Treatment of Diseases

A new chapter in the diagnosis of infectious diseases has been opened by DNA technology, in particular the use of PCR and labeled nucleic acid probes to track down pathogens. For example, because the sequence of the RNA genome of SARS-CoV-2 is known, RT-PCR can be used to amplify and therefore detect and quantify even a small amount of SARS-CoV-2 RNA in nasal or throat swab samples (see Figure 20.11). RT-PCR is often the best way to detect an otherwise elusive infectious agent.

Medical scientists can now diagnose hundreds of human genetic disorders by using PCR with primers that target the genes associated with these disorders. The amplified DNA product is then sequenced to reveal the presence or absence of the disease-causing mutation. Among the genes for human diseases targeted in this way are those for sickle-cell disease, hemophilia, cystic fibrosis, Huntington's disease, and Duchenne muscular dystrophy. Individuals with such diseases can often be identified before the onset of symptoms, even before birth (see Figure 14.19). PCR can also be used to identify symptomless carriers of potentially harmful recessive alleles as part of genetic counseling (see Concept 14.4).

Personal Genome Analysis

As you learned earlier, genome-wide association studies have pinpointed SNPs (single nucleotide polymorphisms) that are linked to disease-associated alleles (see Figure 20.15). Individuals can be tested by PCR and sequencing for a SNP that is correlated with the abnormal allele. The presence of particular SNPs is correlated with increased risk for particular adverse health conditions such as heart disease, Alzheimer's disease, and some types of cancer.

Direct-to-consumer genome analysis companies offer kits allowing individuals to send in a swab containing cheek cells that the company will analyze genetically. Genetic testing for risk factors like heart disease, Alzheimer's disease, and some types of cancer is carried out by looking for linked SNPs that have been previously identified (see Figure 20.15). It may be helpful for individuals to learn about their health risks, with the understanding that such genetic tests merely reflect correlations and do not make predictions.

In addition to health-related genetic information, these companies compare a person's DNA segments with those from reference populations around the world, established from thousands of individuals of known ancestry. Based on how closely the segments match up, the report can tell an individual about their likely ancestral breakdown. Females can also learn about their maternal lines of descent, based on comparisons of mitochondrial DNA, which is contributed to the egg only by the mother. For males, an analysis of the sequence of the Y chromosome can trace their paternal lines of descent. As the size of the database used by these companies increases, the results become more refined and more accurate.

Personalized Medicine

The techniques described in this chapter have also prompted improvements in disease treatments—for example, by knowing which specific cancer-related genes have been mutated in a particular patient's tumor. By analyzing the expression of many genes in large numbers of breast cancer patients, researchers can refine their understanding of the different subtypes of breast cancer (see Figure 18.27). Knowing the expression levels of particular genes in a given individual can help physicians determine the likelihood that the cancer will recur, thus helping them design an appropriate treatment. Given that some low-risk patients have a 96% survival rate over a ten-year period with no treatment, gene expression analysis allows doctors and patients access to valuable information when they are considering treatment options.

Many researchers envision a future of **personalized medicine**, a type of medical care in which each person's specific genetic profile can provide information about diseases or conditions for which the person is especially at risk and help make health-care decisions. As we will see later in the chapter, a *genetic profile* is currently taken to mean a set of genetic markers such as SNPs. Ultimately, however, it will likely mean the complete DNA sequence of an individual—after sequencing becomes inexpensive enough. In fact, a 2016

study suggested that for difficult-to-diagnose patients it might be faster and more economical to sequence just the expressed regions (the exons, together called the *exome*) or even all of the genome right away, rather than running through a standard series of diagnostic tests. The ability to sequence a person's genome quickly and inexpensively is advancing rapidly, perhaps faster than development of appropriate treatments for the conditions. Still, the identification of genes involved in these conditions provides targets for therapeutic interventions.

An individual's genomic information can be used to predict the benefits and risks of particular medications, an approach called *pharmacogenetics*. There are over 300 medications for which the Food and Drug Administration recommends genetic testing for patients prior to their use.

Human Gene Therapy and Gene Editing

Gene therapy—the introduction of genes into an afflicted individual for therapeutic purposes—holds great potential for treating the relatively small number of disorders traceable to a single defective gene. The aim of this approach is to insert a normal allele of the defective gene into the somatic cells of the tissue affected by the disorder.

BBC Video

Cystic Fibrosis and the Promise of Gene Therapy

image

image

For gene therapy of somatic cells to be permanent, the cells that receive the normal allele must be cells that multiply throughout the patient's life. Bone marrow cells, which include the stem cells that give rise to all the cells of the blood and immune system, are prime candidates. **Figure 20.22** outlines one procedure for gene therapy in an individual whose bone marrow cells do not produce a vital enzyme because of a single defective gene. One type of severe combined immunodeficiency (SCID) is caused by this kind of defect. If the treatment is successful, the patient's bone marrow cells will begin producing the missing protein, and the patient may be cured.

Figure 20.22

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Gene therapy using a retroviral vector.
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Insert RNA version of normal allele into retrovirus. Let virus infect bone marrow cells. Viral DNA carrying the normal allele inserts into chromosome. Engineered cells are injected into the patient.

Cloned gene shown (normal allele, absent from patient's cells).

Step 1: Insert RNA version of normal allele into retrovirus or other viral vector. Viral RNA formed in the viral capsid.

Step 2: Let virus infect bone marrow cells that have been removed from the patient and cultured.

Step 3: Viral DNA carrying the normal allele inserts into chromosome.

Step 4: Engineered cells are injected into the patient.

A retrovirus that has been rendered harmless is used as a vector in this procedure, which exploits the ability of a retrovirus to insert a DNA transcript of its RNA genome into the chromosomal DNA of its host cell (see Figure 19.9). If the foreign gene carried by the retroviral vector is expressed, the cell and its descendants will possess the gene product. Cells that reproduce throughout life, such as bone marrow cells, are ideal candidates for gene therapy.

HHMI BioInteractive Video

Gene Therapy

image

image

HHMI BioInteractive Video

The Search for a Mutated Gene

image

image

HHMI BioInteractive Video

Genes as Medicine

image

image

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The procedure shown in Figure 20.22 was used in gene therapy trials for SCID in France in 2000, resulting in the first indisputable success of gene therapy. However, three patients subsequently developed leukemia, a type of blood cell cancer, and one of them died. Researchers have concluded it is likely that the insertion of the retroviral vector occurred near a gene that triggers the proliferation of blood cells. Using a viral vector that does not come from a retrovirus, clinical researchers have treated several other genetic diseases somewhat successfully with gene therapy, including a type of progressive blindness (see Concept 50.3), a degenerative disease of the nervous system, and a blood disorder involving the β-globin gene.

Gene therapy raises many technical issues. For example, how can the activity of the transferred gene be controlled so that cells make appropriate amounts of the gene product at the right time and in the right place? How can we be sure that the insertion of the therapeutic gene does not harm some other necessary cell function? As more is learned about DNA control elements and gene interactions, researchers may be able to answer such questions.

A more direct approach that avoids the complications of using a viral vector in gene therapy is made possible by gene editing, especially using the CRISPR-Cas9 system (see Figure 17.28). In this approach, the existing defective gene is edited to correct the mutation.

HHMI BioInteractive Video

Jennifer Doudna: CRISPR Provides a Tool for Changing Genes

image

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Researchers have been using the CRISPR-Cas9 system to try to correct the genetic mutation that leads to sickle-cell disease. In 2018, they reported that they had injected gene-edited human cells from patients with sickle-cell disease into the bone marrow of mice, with promising results. Employing an improved technique that uses CRISPR, a similar experiment showed that, 16 weeks later, the gene was altered appropriately in 70% of the cells. This is exciting, as it suggests that CRISPR technology could potentially treat or even cure human genetic diseases, such as sickle-cell, Parkinson's, or Alzheimer's disease, or even some types of cancer.

There are ongoing concerns about introducing CRISPR-treated cells into humans because of the risks associated with cutting both strands of DNA and the possible effects on so-called "off-target genes," genes that are not meant to be edited. Researchers are studying ways of minimizing these effects. One approach that has been successful is to use CRISPR in association with a new technique called *base editing*, in which

specific bases in a gene are chemically modified to resemble a different base, rather than being cut and replaced. This is done in a way that alters the resulting protein so it is no longer harmful.

In addition to technical challenges, gene therapy and gene editing provoke ethical questions. Some critics believe that tampering with human genes in any way is unethical. Other observers see no fundamental difference between transplanting genes into somatic cells and transplanting organs from one person to another.

An even more pressing question is the issue of engineering human germ-line cells to try to correct a defect in future generations. Such genetic engineering is now routinely done in laboratory mice, and some methods for genetic engineering of human embryos have been developed.

The development of the CRISPR-Cas9 system has engendered much debate about the ethics of gene editing. Jennifer Doudna (Figure 20.14), a co-discoverer of CRISPR-Cas9, and other biologists have agreed to exercise extreme caution as the field moves forward. Together, they called for the research community to “strongly discourage” any experimental work on human eggs or embryos.

In spite of this general consensus among biologists, a Chinese researcher reported in 2018 that he had used the CRISPR-Cas9 system to edit genes in embryos that completed fetal development and were born as twins and a third individual. He claimed to have edited the CCR5 gene, which codes for a co-receptor for HIV (see Figure 7.8), so that HIV would be unable to bind and infect the cells. The twins’ father was HIV-positive, which was the rationale the scientist used for carrying out this genetic engineering. However, this act was roundly condemned as highly unethical by the biological community; the researcher lost his job and was released in 2022 after 3 years in prison. In 2019, the World Health Organization established the Human Genome Editing Registry to monitor such research.

There are many potential technical problems with the CRISPR-Cas9 technique, like the off-target effects mentioned previously; there is strong evidence that off-target editing did occur in the three gene-edited individuals, although the consequences of these events are unknown at present. Beyond those technical concerns, however, are underlying ethical considerations: Under what circumstances, if any, should genomes of human germ lines be altered? Would alterations lead to the practice of eugenics, a deliberate effort to control the genetic makeup of human populations? It is imperative to consider these questions.

Pharmaceutical Products

The pharmaceutical industry derives significant benefit from advances in DNA technology and genetic research, applying them to the development of useful drugs to treat diseases. Pharmaceutical products are synthesized using methods of either organic chemistry or biotechnology, depending on the nature of the product.

Synthesis of Small Molecules for Use as Drugs

Determining the sequence and structure of proteins crucial for tumor cell survival has led to the identification of small molecules that combat certain cancers by blocking the function of these proteins. One drug, imatinib (trade name Gleevec), is a small molecule that inhibits one tyrosine kinase (see Figure 11.8). The overexpression of this kinase, resulting from a chromosomal translocation, is instrumental in causing chronic myelogenous leukemia (CML; see Figure 15.16). Patients in the early stages of CML who are treated with imatinib have exhibited nearly complete, sustained remission from the cancer. Drugs that work in a similar way have also been used with success to treat a few types of lung and breast cancers. This approach is feasible only for cancers whose molecular basis is fairly well understood.

HHMI BioInteractive Video

Gleevec

image

image

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In many cases of such drug-treated tumors, though, cells later arise that are resistant to the new drug. In one study, the whole genome of the tumor cells was sequenced both before and after the appearance of drug resistance. Comparison of the sequences showed genetic changes that allowed the tumor cells to “get around” the drug-inhibited protein. Here, we can see that cancer cells demonstrate the principles of evolution: Certain tumor cells have a random mutation that allows them to survive in the presence of a particular drug, and as a consequence of natural selection in the presence of the drug, these are the cells that survive and reproduce.

Protein Production in Cell Cultures

Pharmaceutical products that are proteins are commonly synthesized on a large scale using cell cultures. You learned earlier in the chapter about DNA cloning and gene expression systems for producing large quantities of a chosen protein that is present naturally in only minute amounts. The host cells used in such expression systems can even be engineered to secrete a protein as it is made, thereby simplifying the task of purifying it by traditional biochemical methods.

Among the first pharmaceutical products manufactured in this way were human insulin and human growth hormone (HGH). Some 2 million people with diabetes in the United States depend on insulin treatment to control their disease. Human growth hormone has been a boon to children born with a form of dwarfism caused by inadequate amounts of HGH, as well as helping patients with AIDS gain weight. Another

important pharmaceutical product produced by genetic engineering is tissue plasminogen activator (TPA). If administered shortly after a heart attack, TPA helps dissolve blood clots and reduces the risk of subsequent heart attacks.

Protein Production by “Pharm” Animals

In some cases, instead of using cell systems to produce large quantities of protein products, pharmaceutical scientists can use whole organisms. To do this, they use a **transgene**, a gene that has been transferred into one organism from another. They first remove eggs from a female of the recipient species and fertilize them *in vitro*. Meanwhile, they clone the desired gene from a donor organism. They then inject the cloned DNA directly into the nuclei of the fertilized eggs. Some of the cells integrate the foreign DNA, the transgene, into their genome and are able to express the foreign gene. The engineered embryos that arise from these zygotes are then surgically implanted in a surrogate female. If an embryo develops successfully, the result is a **transgenic** organism that expresses its new, “foreign” gene.

Assuming that the introduced gene encodes a protein desired in large quantities, transgenic animals can act as pharmaceutical “factories.” For example, a transgene for a human blood protein such as antithrombin, which prevents blood clots, can be inserted into the genome of a goat in such a way that the transgene’s product is secreted in the animal’s milk **(Figure 20.23)**. The protein is then purified from the milk (which is easier than purification from a cell culture). Such proteins must be tested to ensure that they (or contaminants from the farm animals) will not cause allergic reactions or other adverse effects in patients who receive them.

Figure 20.23

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Goats as “pharm” animals.
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A goat is milked in a laboratory setting.

This transgenic goat carries a gene for a human blood protein, antithrombin, which she secretes in her milk. Patients with a rare hereditary disorder in which this protein is lacking suffer from formation of blood clots in their blood vessels. Easily purified from the goat’s milk, the protein is used to prevent blood clots in these patients during surgery or childbirth.

Forensic Evidence and Genetic Profiles

In violent crimes, body fluids or small pieces of tissue may be left at the scene or on the clothes or other possessions of the victim or assailant. If enough blood, semen, or tissue is available, forensic laboratories can

determine the blood type or tissue type by using antibodies to detect specific cell-surface proteins. However, such tests require fairly fresh samples in relatively large amounts. Also, because many people have the same blood or tissue type, this approach can only exclude a suspect; it cannot provide strong evidence of guilt.

DNA testing, on the other hand, can identify the guilty individual with a high degree of certainty because the DNA sequence of every person is unique (except for identical twins). Genetic markers that vary in the population can be analyzed for a given person to determine that individual's unique set of genetic markers, or **genetic profile**. (This term is preferred over "DNA fingerprint" by forensic scientists, who want to emphasize the heritable aspect of these markers rather than the fact that they produce a pattern on a gel that, like a fingerprint, is visually recognizable.) The FBI started applying DNA technology to forensics in 1988, using a method involving gel electrophoresis and nucleic acid hybridization to detect similarities and differences in DNA samples. This method required much smaller samples of blood or tissue than earlier methods—only about 1,000 cells.

Animation

Genetic Profiles

image

Today, forensic scientists use an even more sensitive method that takes advantage of variations in length of genetic markers called **short tandem repeats (STRs)**. These are tandemly repeated units of two- to five-nucleotide sequences in specific regions of the genome. The number of repeats present in these regions is highly variable from person to person (polymorphic); even for a single individual, the two alleles of an STR may differ from each other. For example, one individual may have the sequence ACAT repeated 30 times at one genome locus and 15 times at the same locus on the other homolog, whereas another individual may have 18 repeats at this locus on each homolog. (These two genotypes can be expressed by the two repeat numbers: 30,15 and 18,18.) PCR is used to amplify particular STRs, using sets of primers that are labeled with colored fluorescent tags; the length of the region, and thus the number of repeats, can then be determined by electrophoresis. The PCR step allows use of this method even when the DNA is in poor condition or available only in minute quantities: A tissue sample containing as few as 20 cells can be sufficient.

In a murder case, for example, this method can be used to compare DNA samples from the suspect, the victim, and a small amount of blood found at the crime scene. The forensic scientist tests only a few selected portions of the DNA—usually 13 STR markers. However, even this small set of markers can provide a forensically useful genetic profile because the probability that two people (who are not identical twins) would have exactly the same set of STR markers is vanishingly small. The Innocence Project, a nonprofit organization dedicated to overturning wrongful convictions, uses STR analysis of archived samples from

crime scenes to revisit old cases. As of early 2022, according to that group, 375 innocent people have been released from prison in the United States as a result of DNA-based forensic and legal work **(Figure 20.24)**.

Figure 20.24

STR analysis used to release an innocent man from prison.

Table compares STR markers for semen samples from two men, showing that one man is innocent and the other is guilty of a crime.

Source of sample	S T R marker 1	STR marker 2	STR marker 3
Semen on victim	17,19	13, 16	12, 12
Earl Washington	16, 18	14, 15	11, 12
Kenneth Tinsley	17,19	13, 16	12, 12

The parts are as follows.

Part a. In 1984, Earl Washington was convicted and sentenced to death for the 1982 rape and murder of Rebecca Williams. His sentence was commuted to life in prison in 1993 due to new doubts about the evidence. In 2000, STR analysis by forensic scientists associated with the Innocence Project showed conclusively that he was innocent. A photo shows Washington just before his release in 2001, after 17 years in prison.

Part b. In STR analysis, selected STR markers in a DNA sample are amplified by PCR, and the PCR products are separated by electrophoresis. The procedure reveals how many repeats are present for each STR locus in the sample. An individual has two alleles per STR locus, each with a certain number of repeats. This table shows the number of repeats for three STR markers in three samples: from semen found on the victim, from Washington, and from another man (Kenneth Tinsley), who was in prison because of an unrelated conviction. These and other STR data (not shown) exonerated Washington and led Tinsley to plead guilty to the murder. The table is as follows:

Genetic profiles can also be useful for other purposes. A comparison of the DNA of a mother, her child, and the purported father can conclusively settle a question of paternity. Sometimes paternity is of historical interest: Genetic profiles provided strong evidence that Thomas Jefferson or one of his close male relatives was the male parent of at least one of the children of Sally Hemings, one of the people Jefferson enslaved. Genetic profiles can also identify victims of mass casualties. The largest such effort occurred after the attack on the World Trade Center in 2001; more than 10,000 samples of victims' remains were compared with

DNA samples from personal items, such as toothbrushes, provided by families. Ultimately, forensic scientists succeeded in identifying almost 3,000 victims using these methods.

Just how reliable is a genetic profile? The greater the number of markers examined in a DNA sample, the more likely it is that the profile is unique to one individual. In forensic cases using STR analysis with 13 markers, the probability of two people having identical DNA profiles is somewhere between one in 10 billion and one in several trillion. (For comparison, the world's population is roughly 8 billion.) The exact probability depends on the frequency of those markers in the general population. Information on how common various markers are in different ethnic groups is critical because these marker frequencies may vary considerably among ethnic groups and between a particular ethnic group and the population as a whole. With the increasing availability of frequency data, forensic scientists can make extremely accurate statistical calculations. Thus, despite problems that can arise from insufficient data, human error, or flawed evidence, genetic profiles are now accepted as compelling evidence by legal experts and scientists alike.

Environmental Cleanup

Increasingly, the diverse abilities of certain microorganisms to transform chemicals are being exploited for environmental cleanup. If the growth needs of such microorganisms make them unsuitable for direct use, scientists can now transfer the genes for their valuable metabolic capabilities into other microorganisms, which can then be used to treat environmental problems. For example, many bacteria can extract heavy metals, such as copper, lead, and nickel, from their environments and incorporate the metals into compounds such as copper sulfate or lead sulfate, which are readily recoverable. Genetically engineered microorganisms may become important in both mining (especially as ore reserves are depleted) and cleaning up highly toxic mining wastes. Biotechnologists are also trying to engineer microorganisms that can degrade chlorinated hydrocarbons and other harmful compounds. These microorganisms could be used in wastewater treatment plants or by manufacturers before the compounds are ever released into the environment.

Agricultural Applications

Scientists are working to learn more about the genomes of agriculturally important plants and animals. For a number of years, they have been using DNA technology in an effort to improve agricultural productivity. The selective breeding of both livestock (animal husbandry) and crops has exploited naturally occurring mutations and genetic recombination for thousands of years.

As we described earlier, DNA technology enables scientists to produce transgenic animals, which speeds up the selective breeding process. The goals of creating a transgenic animal are often the same as the goals of

traditional breeding—for instance, to make a sheep with better quality wool, a pig with leaner meat, or a cow that will mature in a shorter time. Scientists might, for example, identify and clone a gene that causes the development of larger muscles (muscles make up most of the meat we eat) in one breed of cattle and transfer it to other cattle or even to sheep. However, health problems are not uncommon among farm animals carrying genes from other species, and modification of the animal's own genes using the CRISPR-Cas9 system will likely emerge as a more useful technique. Animal health and welfare are important issues to consider when genetically altering animals.

Agricultural scientists have already endowed a number of crop plants with genes for desirable traits, such as delayed ripening and resistance to spoilage, disease, and drought. Modifications can also add value to food crops, giving them a longer shelf life or improved flavor or nutritional value. For many plant species, a single tissue cell grown in culture can give rise to an adult plant. Thus, genetic manipulations can be performed on an ordinary somatic cell and the cell then can be used to generate a plant with new traits.

Genetic engineering is rapidly replacing traditional plant-breeding programs, especially for useful traits, such as herbicide or pest resistance, determined by one or a few genes. Crops engineered with a bacterial gene making the plants resistant to an herbicide can grow while weeds are destroyed, and genetically engineered crops that can resist destructive insects reduce the need for chemical insecticides.

The Food and Agriculture Organization of the United Nations has predicted that we will need 70% more food by the year 2050 than the planet is currently producing. One novel approach is being taken by researchers working on the international C4 Rice Project. The common form of rice, a global food staple, uses the C3 form of photosynthesis (see Concept 10.5). The aim of researchers engaged in this project is to genetically engineer a strain of rice that can use C4 photosynthesis, which is more efficient. This research has run into some hurdles, but a gene was identified in 2022 that, when knocked out, increases production in both rice and corn plants by 8–10%. (The gene is a member of the WD40 family; see Figure 21.3.)

Safety and Ethical Questions Raised by DNA Technology

Early concerns about potential dangers associated with recombinant DNA technology and genetic engineering focused on the possibility that hazardous new pathogens might be created. What might happen, for instance, if in a research study cancer cell genes were transferred into bacteria or viruses? To guard against such rogue microorganisms, scientists developed a set of guidelines that were adopted as formal government regulations in the United States and some other countries. One safety measure is a set of strict laboratory procedures designed to prevent engineered microorganisms from either infecting researchers or accidentally leaving the laboratory. In addition, strains of microorganisms to be used in recombinant DNA experiments are genetically crippled to ensure that they cannot survive outside the laboratory. Finally, certain obviously dangerous experiments have been banned.

Today, most public concern about possible hazards centers not on recombinant microorganisms but on **genetically modified organisms (GMOs)** used as food. A GMO is a transgenic organism that has acquired one or more genes from another species or from another variety of the same species. Some salmon, for example, have been genetically modified by addition of a more active salmon growth hormone gene. However, the majority of the GMOs that contribute to our food supply are not animals, but crop plants.

GM crops are widespread in the United States, Argentina, and Brazil; together, these countries account for over 80% of the world's acreage devoted to such crops. In the United States, most corn, soybean, and canola crops are genetically modified, and a recent law requires labeling of GM products, although as of 2022, the term "GMO" has been replaced by "bioengineered." The same foods are an ongoing subject of controversy in Europe, where the GM revolution has been met with strong opposition. Many Europeans are concerned about the safety of GM foods and the possible environmental consequences of growing GM plants. Although a small number of GM crops have been grown on European soil, the European Union established a comprehensive legal framework regarding GMOs in 2015. Among other regulations, individual member states may ban either the growing or importing of GM crops, which must be clearly labeled. As of 2022, 19 of the 27 European Union member states have fully or partially banned GMOs. However, pressure is increasing to reconsider this stance, based on evidence that improving crop yields could significantly reduce carbon emissions and thus decrease climate change. In the end, though, the high degree of consumer distrust in Europe makes the future of GM crops there uncertain.

Advocates of a cautious approach toward GM crops fear that transgenic plants might pass their new genes to close relatives in nearby wild areas. We know that lawn and crop grasses, for example, commonly exchange genes with wild relatives via pollen transfer. If crop plants carrying genes for resistance to herbicides, diseases, or insect pests pollinated wild ones, the offspring might become "super weeds" that are very difficult to control. Another worry involves possible risks to human health from GM foods. Some people fear that the protein products of transgenes might lead to allergic reactions. Although there is some evidence that this could happen, advocates claim that these proteins could be tested in advance to avoid producing ones that cause allergic reactions. (For further discussion of plant biotechnology and GM crops, see Concept 38.3.)

Today, governments and regulatory agencies throughout the world are grappling with how to facilitate the use of biotechnology in agriculture, industry, and medicine while ensuring that new products and procedures are safe. In the United States, such applications of biotechnology must be evaluated for potential risks by various regulatory agencies, including the Food and Drug Administration, the Environmental Protection Agency, the National Institutes of Health, and the Department of Agriculture. Meanwhile, these same agencies and the public must consider the ethical implications of biotechnology.

Advances in biotechnology have allowed us to obtain complete genome sequences for humans and many other species, providing a vast treasure trove of information about genes. We can ask how certain genes differ from species to species, as well as how genes and, ultimately, entire genomes have evolved. (These are the subjects of Chapter 21.) At the same time, the increasing speed and falling cost of sequencing the genomes of individuals are raising significant ethical questions. Who should have the right to examine someone else's genetic information? How should that information be used? Should a person's genome be a factor in determining eligibility for a job or insurance? Ethical considerations, as well as concerns about potential environmental and health hazards, will likely slow some applications of biotechnology. There is always a danger that too much regulation will stifle basic research and its potential benefits. On the other hand, genetic engineering—especially gene editing with the CRISPR-Cas9 system—enables us to profoundly and rapidly alter species that have been evolving for millennia. A good example is the potential use of a gene drive that would eliminate the ability of mosquito species to carry diseases or even eradicate certain mosquito species. There would probably be health benefits to this approach, at least initially, but unforeseen problems could easily arise. Given the tremendous power of DNA technology, we must proceed with humility and caution.

Interview

Interview with David Suzuki: Exploring ethical issues related to DNA technology

David Suzuki.